Three Proofs of the Steinhaus Theorem and some Corollaries

2026

1 Steinhaus Theorem

Theorem 1.1 (Steinhaus)**.** Given a set A with positive Lebesgue measure, the set of differences $A - A$ contains an open interval around the origin.

There are many proofs to this theorem. Here is the most replicable one:

Lemma 1.2 (Bass Exercise 4.10)**.** Given a set A with positive Lebesgue measure, for all $\alpha \in (0,1)$, there exists an open interval I such that

$$\mu(A \cap I) > \alpha\mu(I).$$

Proof of Lemma 1.2. Choose $\epsilon < \frac{(1-\alpha)}{\alpha}\mu(A)$. We can approximate A with an open set O within ϵ, i.e.

$$\mu(O) < \mu(A) + \epsilon.$$

Every open set is a countable disjoint of disjoint open intervals, and by PHP, there exists at least one open interval O_i of measure greater than zero such that A has density at least as big as $\frac{\mu(A)}{\mu(A)+\epsilon}$ in O_i. This means

$$\frac{\mu(A)}{\mu(A)+\epsilon}\mu(O_i) \leq \mu(A \cap O_i).$$

But since

$$\begin{aligned}\frac{\mu(A)}{\mu(A)+\epsilon} &= \frac{\mu(A)}{\mu(A) + \frac{1-\alpha}{\alpha}\mu(A)} \\ &= \alpha,\end{aligned}$$

we have

$$\alpha\mu(O_i) \leq \mu(A \cap O_i)$$

where O_i is an open interval, as desired. □

Proof of Steinhaus. Suppose the set of differences does not contain an interval, that is, for any $\epsilon > 0$ there exists some $0 < \lambda < \epsilon$ such that $A + \lambda \neq A$. By the lemma, let $\alpha = .75$. Then there exists some open interval I such that $\mu(A \cap I) > .75\mu(I)$. Letting $\frac{\mu(I)}{2} > \epsilon > 0$, we see that the same holds for $A + \lambda$ and $I + \lambda$, that is, $\mu(A + \lambda \cap I + \lambda) > .75\mu(I + \lambda)$.

Note that $\mu(A \cup A + \lambda) = \mu(A) + \mu(A + \lambda)$. Then we have that

$$\begin{aligned} .75\mu(I \cup I + \lambda) &= .75\mu(I) + .75\mu(I + \lambda) \\ &= 1.5\mu(I) \\ &< \mu(A \cap I) + \mu(A + \lambda \cap I + \lambda) \\ &= \mu[(A + \lambda \cap I + \lambda) \cup (A \cap I)] \\ &\leq \mu(I \cup I + \lambda) = 1.5\mu(I) \end{aligned}$$

which is a contradiction. □

Motivating Problem 1.3

Show that an additive function $f : \mathbb{R}_{>0} \to \mathbb{R}_{>0}$ which is Lebesgue measurable is linear.

It is easy to see that $f(nx) = nf(x)$ for all integers n, and that $f(x) = nf\left(\frac{x}{n}\right)$ for any n as well. This means $f(mx) = mf(x)$ whenever $m \in \mathbb{Q}$. So f splits $\mathbb{R}$ up into cosets of the form $r\mathbb{Q}$ with $r \in \mathbb{R}$. It is less obvious that we can get homogeneity for irrational scalars.

Lemma 1.4. Every Lebesgue measurable subset of $\mathbb{R}$ with measure greater than zero contains a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$

A Hamel basis is a collection $\mathcal{H}$, such that for any real number r, there exists a finite unique linear combination

$$\sum_{i=1}^{n} q_i e_i = r,$$

where $q_i \in \mathbb{Q}$, and e_i are elements of the Hamel basis. Note that a Hamel basis contains uncountably infinitely many elements, so it is important that any real number can be expressed in finitely many terms.

Proof of Lemma 1.4. Any open interval contains a Hamel basis for $\mathbb{R}$ over $\mathbb{Q}$, since $(-\epsilon, \epsilon)\mathbb{Q} = \mathbb{R}$ for all $\epsilon > 0$. Steinhaus tells us that for any set A of measure greater than zero, any number in such an open interval can be written as a linear combination (more specifically, a difference) of numbers in A. So every Borel set in $\mathbb{R}$ set of measure greater than zero contains a Hamel basis of $\mathbb{R}$ over $\mathbb{Q}$. □

Now we can answer problem 1.3:

Proof. Consider for each $a \in \mathbb{R}$ the set $A_a = \{x : f(x) = ax\}$. Whenever $\mu(A_a) > 0$, we know that it must contain a Hamel basis for $\mathbb{R}$, which implies it must be all of $\mathbb{R}$, since for any finite subset $S \subset A_a$, additivity is the same as closure under $\mathbb{Q}$-linear combinations. So for each a, $A_a = \mathbb{R}$ or $\emptyset$, so there is exactly one a for which $A_a = \mathbb{R}$, so for all $x \in \mathbb{R}$, $af(x) = f(ax)$, so f is linear and we're done. □

Corollary 1.5. Any proper subgroup of $\mathbb{R}^+$ has Lebesgue measure zero.

Proof. If a proper subgroup had Lebesgue measure greater than zero, then it would contain a Hamel basis, and thus would be able to generate all of $\mathbb{R}$. □

Example 1.6. Use Zorn's lemma to construct a function $f : \mathbb{R} \to \mathbb{R}$ which is additive but not linear

Proof. We want to be able to construct a function that is additive on the cosets $r\mathbb{Q}$ but is not linear, which means we want to be able to be certain that when we select different cosets they are not the same, i.e., we want a set of r_i such that $r_i\mathbb{Q} \neq r_j\mathbb{Q}$ for all $i \neq j$.

Take a poset

$$\left\{ S \in \mathbb{P} : \sum_{i=1}^{N} q_i x_i \neq 0, \forall N \in \mathbb{N}, \forall q_i \in \mathbb{Q}, \forall \{x_i\}_{i \in [[0,N]]} \right\}$$

ordered by inclusion. The maximal element (which we can find by Zorn) S can recover any $\mathbb{R}$ from a $\mathbb{Q}$-linear (finite) combination of numbers in S.

A function that sends each $x \in r\mathbb{Q}$ to $a_r x$ to some constant dependent on r is additive but not linear (if some of the a_rs are not equal to each other), so we're done. □

Note that this implies the existence of a non Lebesgue-measurable function.

Theorem 1.7. There exists sets of measure zero which contain a Hamel Basis of $\mathbb{R}$ over $\mathbb{Q}$. Link

Proof. The cantor set is an example. □

This means there exists Hamel Bases which are Lebesgue measurable, since every set of measure zero is Lebesgue measurable.

1.0.1 Alternative Proofs of Steinhaus

Lemma 1.8. Let μ be Lebesgue measure and A be a measurable subset of $\mathbb{R}$. If $\mu(A) > 0$, then there exists a bounded compact subset A' with $\mu(A') > 0$.

Proof of Lemma. We will first show that for any bounded (we can assume this WLOG) set B of measure 0, there exists an open set containing B such that the measure of their difference is less than ϵ.

Since B is Lebesgue measurable, we know that for all $\epsilon > 0$, there exists some collection of right-open intervals $\{[a_i, b_i)\}_{i \in \mathbb{N}}$ with $E_k := \cup_{i=1}^{k} [a_i, b_i)$ such that $B \subseteq E^\infty$ and $\mu(E^\infty - B) > \frac{\epsilon}{2}$. The union of open intervals

$$O = \bigcup_{i=1}^{\infty} (a_i - \frac{\epsilon}{2^{i+3}}, b_i)$$

is clearly an open set containing B which is within ϵ of B.

We can use this to show that we can find a closed set. Take some larger interval containing B, call it I. By what we showed above, for all $\epsilon > 0$ we can find an open set A containing ϵ of $I \backslash B$ within ϵ. The complement of A (with respect to the interval) is a closed and bounded set contained in B, so is compact, as desired.

□

Proof of Steinhaus. It suffices to prove the result for bounded compact subsets of $\mathbb{R}$ with positive measure. Suppose the difference set contains no open interval around 0. Then there is a sequence (x_n) converging to zero which is in $\mathbb{R} \backslash (A - A)$. Since for each i, there are no two points whose difference is x_i, we have that A and $A + x_i$ are disjoint for each i. Link

For any $\epsilon > 0$ want to construct a subsequence $x_{\sigma(i)}$ such that $\{A + x_{\sigma(i)}\}$ are pairwise disjoint and where $|x_{\sigma(i)}| < \epsilon$. We will proceed by strong induction:

1. $x_{\sigma(1)}$ exists because (x_i) converges to 0.

2. Suppose $\{x_{\sigma(i)}\}_{i<k-1}$ are mutually disjoint and within $(-\epsilon, \epsilon)$. Towards contradiction, suppose for every integer $n > \sigma(k-1)$ there exists some i such that $A + x_n \cap \bigcup_{i=1}^{i}(A + x_{\sigma(i)}) \neq \emptyset$.

   Since k is finite, there exists some positive integer $l < k$ for which $x_{\sigma(l)} + A$ intersects infinitely many of the $x_n + A$. Call the subsequence of x_n which do intersect the union $\bigcup_{i=1}^{l} x_{\sigma(i)} + A$, (y_i). So $(y_i - x_{\sigma(l)})_{i\in\mathbb{N}}$ is a sequence contained in D converging to $x_{\sigma(l)}$ which is not contained in D, a violation of sequential compactness, which means we have a contradiction.

   Thus $\sigma(k)$ exists.

So we have a subsequence $x_{\sigma(i)}$ converging to zero contained in $(-\epsilon, \epsilon)$ for which $A + x_{\sigma(i)}$ are all pairwise disjoint. Then the union

$$\bigcup_{i=1}^{M} A + x_{\sigma(i)}$$

is a set contained in $[\inf(A) - \epsilon, \sup(A) + \epsilon]$[1] with measure $M\mu(A)$. Since M can be arbitrarily large for any ϵ, there is a certain point where $M\mu(A) > \ell([\inf(A) - \epsilon, \sup(A) + \epsilon])$, a contradiction.

□

Proof of Steinhaus. Proceeding similarly to the previous proof, suppose A is compact and bounded, Link and find a sequence (x_i) converging to zero which is not contained in the difference set.

Since A is bounded, there exists an open set $U \supset A$ □

Lemma 1.9 (Convolutions are Smoothing)**.** If $|f|$ is integrable, and g is bounded almost everywhere, then the convolution

$$f * g(x) = \int f(y)g(x-y)\,\mathrm{d}y$$

is continuous and bounded.

Proof of Lemma. Since g is bounded a.e., there exists some $M \in \mathbb{R}$ and a set E with $\mu(E) = 0$ such that $g(x) < M$ for all $x \in E^C$. This means

$$\int |f(y)g(x-y)|\,\mathrm{d}y = \int_{E^C} |f(y)g(x-y)|\,\mathrm{d}y = \int_{E^C} |f(y)|M\,\mathrm{d}y = M\int |f(y)|\,\mathrm{d}y < \infty$$

Since

$$f * g(x) < \int |f(y)g(x-y)|\,\mathrm{d}y,$$

we have that $f * g(x)$ is bounded.

The convolution $f * g$ is continuous if for all $z \in \mathbb{R}$,

$$\lim_{x\to z} f \star g(x) = f * g(z)$$

[1] remember A is bounded.

or equivalently

$$\lim_{x \to z} \int f(y)g(x-y)\,\mathrm{d}y = \int f(y)g(z-y)\,\mathrm{d}y.$$

We showed at the beginning that $|f(y)g(x-y)|$ is integrable, so we can apply the dominated convergence theorem and we see that $f * g$ is continuous. $\square$

Proof of Steinhaus. Reduce this problem to the case where A is bounded. By the lemma, we see that $\mathbb{1}_A * \mathbb{1}_{-A}$ must be a continuous and bounded function. Notice that Link

$$\mathbb{1}_A * \mathbb{1}_{-A} = \int_{\mathbb{R}} \begin{cases} 1 \text{ if } y, y-x \in A \\ 0 \text{ otherwise} \end{cases} \mathrm{d}y$$

So this function is greater than zero at x if there is a set $S \subseteq A$ with $\mu(S) > 0$ such that $S - x \subseteq A$ as well. Then clearly $\mathbb{1}_A * \mathbb{1}_{-A}(0) = \mu(A)$, and since the function is continuous there must exist some open interval around 0 where it is positive, and the result follows. $\square$